AGRICULTURAL DRAINAGE · NOTE DR02

Drainage Investigations — Factors, Benefits, Wells, Piezometers, Maps and Drainage Coefficient

Course topics: IDE-502 T5, IDE-502 T6, CAE1.3 L14, IDE-515 T12

Factors influencing drainage

Factor	Influence
Topography	Sets the plan and the locations of the outlet and the field, link and main drains. Good surface slopes clear runoff quickly and reduce the need for subsurface drainage. Flat irrigated land — especially in arid regions — often has no natural outlet, so drainage water may have to be pumped to a stream or river
Soil	Artificial drainage is needed when a high water table must be lowered or surface water cannot move down or away fast enough. Permeable soils rarely need it (unless the water table is high); slowly permeable soils often do, especially on level land with high rainfall. Finer texture → smaller pores → greater need: it falls from clay → silt → loam → sandy loam → sand
Rainfall	Produces runoff and infiltration; excess water must leave quickly. Surface and subsurface water usually head for the same outlet (the topographic low), so design for the total disposal capacity
Crop	Sets the tolerable depth and duration of wetness

- **Structure:** platy structure drains poorly; prismatic allows better downward movement; blocky and granular structures favour drainage.
- Surface infiltration and the permeability of lower horizons matter most; hard pans and barriers cause waterlogging. Coarse soils drain better than fine ones (though not always in line with permeability).
- Irrigated soils are often layered — find the coarse, drainable layers by boring; the sequence of permeable and impermeable layers decides the type and design of the system.
- **Subsoil colour:** uniform colour — usually no problem; yellow — intermediate permeability; red, yellow and grey mottling — slower permeability; grey clay in humid regions — very slow.

Design storm	Return period
Surface drains	5–15 years
Major cross-drainage works, embankments, canals	25 years
Minor crossings	10 years
Field structures	5 years

Ponding on arable land should not last beyond about 48 hours, or production falls and crops may be destroyed.

Land that needs drainage

- A high water table near the root zone — within 1.5 m is undesirable.

- Standing surface water for long periods — tolerable only 2–4 hours for sensitive vegetables, about 1 day for maize and small grains, 4 days or more for water-tolerant grasses.
- Regions where evaporation exceeds rainfall (salinity, alkalinity, rising groundwater); soil moisture above field capacity for long periods; humid regions with continuous or intermittent heavy rain; flat land with fine-textured soils.

Choosing the type of system

- Surface runoff collecting → **surface drainage**. Shallow water table → **subsurface drainage**. Where conventional drainage is unfeasible or uneconomic → **vertical drainage** (wells) or **bio-drainage** (high water-use trees).
- Provide surface drainage from the start of canal irrigation to prevent waterlogging and salt build-up; on land already waterlogged or salt-affected, subsurface drainage is best for reclamation.

Benefits of drainage

- A better root environment: deeper rooting and better extraction of nutrients and water; aeration restored, with beneficial microbes and nutrient availability; better structure and infiltration, so less erosion.
- Better tillage; easier interculture and harvest; the soil warms faster, allowing earlier planting, better germination and growth and a longer season.
- Salts leached for higher productivity; more nitrogen (denitrification prevented); toxic substances and disease organisms of wet soils removed.
- Less lodging (deeper roots, firmer soil); wider crop choice; easier weed control; more productive, less compacted grazing land; fewer diseases of plants, animals and people (root rot, stem rot, foot rot, liver fluke, mosquito-borne disease).

Drainage investigations

- **Sources of waterlogging:** seepage from canals, hillsides or higher irrigated land; runoff collecting in low land without an outlet; canal irrigation of level land raising the water table and salinity; over-irrigation beyond the drainage capacity; seepage from surrounding unirrigated hillsides into valleys.
- About 1 cm of excess percolating water can raise the water table by 8–10 cm. Canal seepage in permeable soils may be 10–30% of the water delivered, sometimes appearing 5–10 km away.
- **Purpose:** identify the sources and extent of waterlogging, the system needed for optimum growth, and its techno-economic feasibility.
- **General study:** outlet location, runoff from the watershed, rainfall, severity of the problem, soils, crops and land value — followed by topographic and groundwater surveys, runoff characteristics and drain locations, and preliminary costs.

Instrument	Construction and use
Observation well	A hole to watch water-table fluctuations — its water level is the water table at that moment. Usually 2.5 cm G.I. pipe, screened through the range of the expected maximum water table, with ≈ 3 mm perforations giving ≈ 20% open area. In fine sand, gravel-pack it — e.g. a 5 cm auger hole for a 2.5 cm pipe, with gravel that retains ≈ 90% of the formation material; a coir strainer may serve in coarse formations
Piezometer	A small vertical pipe (≈ 2.5 cm) open only at the bottom, measuring the hydrostatic pressure of groundwater at that point. Install on a grid 30–300 m or more apart, in batteries at different depths at least 60 cm apart. Drive it with a protected driving head, its lower end plugged and then opened at depth (remove the soil inside with a screw auger); flush with a portable piston pump to form a ≈ 15 cm cavity at the bottom, about every 6 months against clogging — or install by jetting with a high-pressure, low-volume pump

- **Readings:** reference the ground RL to a permanent bench mark; measure water depth with an electric depth gauge or chalked tape — daily in the rainy season if needed, otherwise weekly, fortnightly or monthly. They give the water-table depth, the direction and the location of groundwater movement.
- **Vertical gradient (piezometer battery):** deeper level higher than shallow → upward flow; lower → downward flow; equal → no vertical flow. Batteries at different places give the horizontal flow direction.

$$\text{Hydraulic gradient} = [(h_1 + Z_1) - (h_2 + Z_2)] \,/\, (Z_2 - Z_1)$$

h = pressure head (water column above the piezometer tip) and Z = elevation of the tip, so h + Z is the hydraulic head at each tip; divide the head difference by the vertical distance between the tips. (The bracket around the numerator is missing in the notes.)

Map	Shows and uses
Water-table contour map	Lines of equal hydraulic head — the water-table surface at a given time, its slope and the direction of flow. Steep slopes mean barriers or low conductivity; gentle slopes a permeable aquifer with horizontal flow
Isobath map	Lines of equal depth to water table (more points where slope changes fast) — delineates the area needing drainage and evaluates measures
Observation-well hydrograph / profile	Water-surface elevation along a line — flow direction is normal to the equal-head lines

- **Soil chemistry:** salinity, ESP or SAR and pH of representative samples to 1.5 m (root-zone depth) over the whole problem area, analysed by standard methods. Feasibility also depends on productivity after drainage, the soil and subsoil, and an adequate outlet — often needing several farms or a whole watershed to cooperate.
- **Survey information needed:** watershed map, vegetation, outlet ditch, roads, railways and features; profiles of proposed ditches and tile mains; topographic map with low-point elevations; outlet conditions (capacity, floods, water level); soil properties, infiltration and permeability; soluble salts to be leached; exchangeable sodium.

Drainage coefficient

- The depth of water to be drained from an area in 24 hours (or the flow per unit area) — the design drainage rate.
- It depends on location, land use and area, rainfall amount, intensity and duration, climate, soil permeability and crop. Rainfall dominates in humid areas; irrigation and field losses in arid and semi-arid irrigated areas.
- Typically 6 mm to 25 mm for small projects, based on experience under similar soil, crop and climate.

Sources: ICAR eCourse 'Drainage Engineering' — Dr M. K. Jha (IIT Kharagpur), reviewed by Prof. K. Yellareddy (Director A&R, WALAMTARI, Hyderabad); AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. About 1 cm of excess percolating water can raise the water table by

(a) 1 cm (b) 3–4 cm
(c) 8–10 cm (d) 50 cm

Q2. A piezometer admits water

(a) along its whole length (b) only at its open bottom
(c) through a screen (d) from the surface

Q3. A map of equal depths to the water table is an

(a) isohyet (b) isobath
(c) isochrone (d) isopleth

Q4. Typical drainage coefficient for small projects is

(a) 0.6–2.5 mm/day (b) 6–25 mm/day
(c) 60–250 mm/day (d) 1 m/day

Q5. Ponding on arable land should not exceed about

(a) 6 h (b) 24 h
(c) 48 h (d) 7 days

Q6. Need for artificial drainage is greatest in

(a) sand (b) sandy loam
(c) loam (d) clay

Answers

Q1 — (c) 8–10 cm — *Specific yield ~10–12%.*
Q2 — (b) only at its open bottom
Q3 — (b) isobath
Q4 — (b) 6–25 mm/day
Q5 — (c) 48 h
Q6 — (d) clay